

ERRORS AND UNCERTAINTIES

1. TYPES OF ERRORS

We discern three types of errors:

- *Gross errors*: Making a wrong reading of a value on a meter stick is a typical gross error. Gross errors can be avoided by careful experimentation.
- *Systematic errors*: Systematic errors are caused by imprecise measuring meters or methods, e.g. a meter stick which is slightly too long. Systematic errors bias all measured values in the same direction and cannot be eliminated by repeated measurements. They are usually very difficult to detect.
- *Random errors*: There is no absolutely precise measurement. Even the most sophisticated meters have only a limited precision. In electronic meters, the electronic “noise” caused by the thermal motion of the electrons sets a lower limit on precision. Since random errors do not change the “correct” result in any preferred direction, their influence can be reduced by repeated measurements.

2. RULE OF THUMB

The result of a calculation can have at most as many significant figures as the least accurate quantity used in the calculation.

The rule of thumb only gives a rough idea of the result’s accuracy.

3. ABSOLUTE AND RELATIVE UNCERTAINTIES

An uncertainty is an estimate of the range within which a measured value lies near the true value. The *absolute* uncertainty of a measured value a is Δa , if a is in the interval $[a - \Delta a, a + \Delta a]$ with absolute certainty*. We write the result as $a \pm \Delta a$, e.g. $N_A = (6.02 \pm 0.02) \cdot 10^{23} \text{ mol}^{-1}$. In this example, there is no point in writing the quantity as $N_A = 6.0221367 \cdot 10^{23} \text{ mol}^{-1}$ since the value of the last five figures is smaller than the uncertainty.

We always round the uncertainty to one significant figure† and the result to the same decimal place. Without explicitly giving an uncertainty, we imply it to be of the order of the last figure of the number. For a physicist, a value of 1.5 is *not* the same as 1.500!

Example: The result of a calculation is 153.173 ms and the calculated uncertainty $3.21 \cdot 10^{-4} \text{ s}$. The standard form of the result is therefore $(153.2 \pm 0.3) \text{ ms}$.

The *relative* uncertainty of a measured value a is the ratio of the absolute uncertainty and the value itself:

$$r_a = \frac{\Delta a}{a}.$$

Relative uncertainties are often expressed as a percentage, e.g. $N_A = 6.0221367 \cdot 10^{23} \text{ mol}^{-1} \pm 0.3 \%$.

4. ERROR PROPAGATION

Only rarely does an experiment give us directly the quantities we are interested in. Instead, we calculate them from measured values. The uncertainty of the result can be calculated from the uncertainties of the measured quantities using the method of maximum uncertainty:

Calculate the worst result you can expect taking the uncertainties of the underlying values into account. The difference to the expected result corresponds to the absolute uncertainty.

For “small” relative uncertainties we can apply the following rules for *error propagation* (see „Formulae, Tables, Concepts“, p 184):

- Addition and subtraction: Absolute uncertainties are added.
- Multiplication and division: Relative uncertainties are added.
- Powers: Relative uncertainties are multiplied by the power.

* In addition to absolute (maximum) uncertainties, it is quite common in physics to use standard uncertainties. Typically, the standard uncertainty corresponds to one standard deviation.

† As an exception to the rule a value starting with the (significant) figure 1 is often rounded to two significant figures.

5. DIAGRAMS

In diagrams, measured values are represented by dots or small boxes and by *error bars* (see fig. 1).

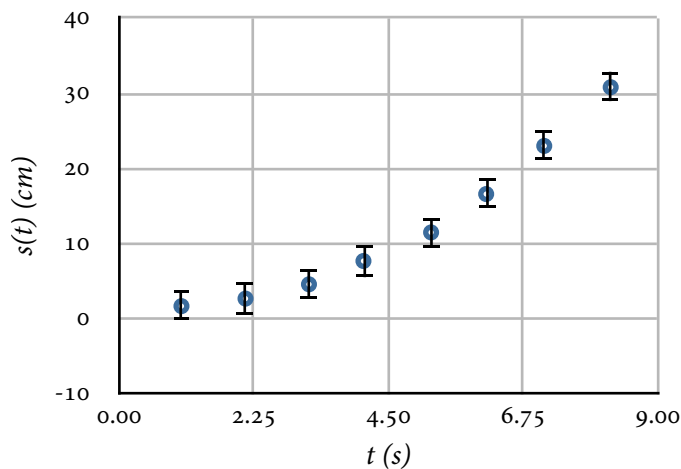


FIGURE 1: Diagram with error bars

If there is a theoretical prediction for the relation between the two quantities, we can fit a function to the data points (*regression*) using an appropriate tool (e.g. SciPy's *curve_fit* method). The hypothesis is verified if the regression line crosses most of the error bars.

The quality of a curve fit is often verified by means of a *residual analysis*. A residual plot displays the difference between calculated and measured value (i.e. the *residuals*) including error bars. For a good fit the residuals should be randomly distributed about zero and the error bar should include the horizontal axis.

6. EXAMPLES

a) Finding absolute uncertainty

- Measuring lengths using a meter stick with millimeter division: Readings can be made to an accuracy of at least 0.5 mm.
- Measuring time using a stop watch with division 0.01 s: The response time of a human and therefore the absolute uncertainty is of the order 0.1 s. Do not be fooled by digital displays! The accuracy can be improved by repeated measurements of the same event.

b) The speed of a vehicle can be determined by measuring the time required for it to cover a certain distance. The measured values are $t \pm \Delta t$ and $s \pm \Delta s$. The speed is $v = s/t$.

Using the method of maximum uncertainty, the worst result can be calculated as $v_{\max} = (s + \Delta s)/(t - \Delta t)$. The uncertainty for the speed therefore is $\Delta v = v_{\max} - v$.

Alternatively the quotient rule can be used: The relative uncertainties of distance and time have to be added: $r_v = r_s + r_t = \Delta s/s + \Delta t/t$.

The absolute uncertainty of the speed then is $\Delta v = v \cdot r_v = v \cdot \left(\frac{\Delta s}{s} + \frac{\Delta t}{t} \right)$.